

Econ 102 HW 1

1)

a)

Japan			USA		
Exp	Income	Production	Exp	Income	Production
C 50	CP 100	SD sales	C 50	CP 100	SD sales
I 50	I	DB saved	I 50	I	SD saved
G 0	D		G 0	D	
X 100	W		X 100	W	
-M 100	R		-M 100	R	
100	100	100	100	100	100

Exp	Income	Production
C 0	CP 2000	Amun: 3000
I 0	I 0	Invest: 10000
G + 5000 + 10000	D 0	
X 0	W 10,000 + 10000	
-M 2000	R 0	
13000	17000	13000

Exp	Income	Prod	Exp	Income	Prod
C 10000 + 2000	CP 10000 + 20000	220 000 Jans	C 5000 + 7000	CD 15000 + 2000	Jans 7000
I 10000	I 0		I 50000	I	
G 0	D 0		G	G	
X 0	W 0		X	X	
-M 0	R 0		-M	R	
220 000	220 000	220 000	170000	170000	170000

d) At first, 0 contribution. After we incorporate, there is still 0 contribution because there are no transactions in the market.

E) This transaction is included if the house being purchased is new or it was built within the last year. Both are investments for Exp Method.

d)

$$a) \text{ Auto: } 10 \cdot 10 + 9 \cdot 10 + 10 \cdot 10 = 290$$

$$\text{Home: } 10 \cdot 12 + 11 \cdot 20 + 20 \cdot 15 = 640$$

$$\text{Entertainment: } 12 \cdot 5 + 5 \cdot 6 + 10 \cdot 6 = 150$$

$$2019_{\text{new}}: 10 \cdot 10 + 10 \cdot 12 + 12 \cdot 5 = 280$$

$$2020: 9 \cdot 10 + 11 \cdot 20 + 5 \cdot 6 = 340$$

$$2021: 10 \cdot 10 + 20 \cdot 15 + 10 \cdot 6 = 460$$

b) Between 2019 and 2021, Automobiles were constant, house prices went way up due to the pandemic (hard to be at home most), and Entertainment dipped in 2020 when people stopped going outside for recreational post-pandemic.

c) 2021 base

$$\begin{array}{r} 2019_{\text{new}}: 10 \cdot 10 \\ + 20 \cdot 12 \\ + 10 \cdot 5 \\ \hline = 340 \end{array}$$

2020 base

$$\begin{array}{r} 2019_{\text{new}}: 9 \cdot 10 \\ + 11 \cdot 12 \\ + 5 \cdot 5 \\ \hline = 247 \end{array}$$

2019 base

$$2019_{\text{new}} = 280$$

$$\begin{array}{r} 2020_{\text{new}}: 10 \cdot 10 \\ + 20 \cdot 20 \\ + 10 \cdot 6 \\ \hline = 560 \end{array}$$

$$2020_{\text{new}} = 340$$

$$\begin{array}{r} 2020: 10 \cdot 10 \\ + 10 \cdot 20 \\ + 12 \cdot 6 \\ \hline = 372 \end{array}$$

$$2021_{\text{new}} = 460$$

$$\begin{array}{r} 2021_{\text{new}}: 9 \cdot 10 \\ + 11 \cdot 15 \\ + 5 \cdot 6 \\ \hline = 285 \end{array}$$

$$\begin{array}{r} 2021: 10 \cdot 10 \\ + 10 \cdot 15 \\ + 12 \cdot 6 \\ \hline = 322 \end{array}$$

There are big changes due to changing prices/quantities in these goods.

d) we use chained prices so that inflation doesn't impact year to year changes

3)

$$\begin{array}{r} 1950 \\ 40 \cdot 10 \\ \hline 8 \cdot 10 \\ \hline 480 \end{array}$$

$$\begin{array}{r} 1980 \\ 40 \cdot 20 \\ \hline 8 \cdot 12 \\ \hline 896 \end{array}$$

$$\begin{array}{r} 2024 \\ 40 \cdot 40 \\ \hline 8 \cdot 14 \\ \hline 1600 \end{array}$$

$$1950 \rightarrow 80: \frac{800 - 400}{400} = \boxed{100\%}$$

$$1980 \rightarrow 2024: \frac{1600 - 800}{800} = \boxed{100\%}$$

$$\begin{array}{r} 1950 \\ 40 \cdot 10 \\ + 8 \cdot 10 \\ \hline 480 \end{array}$$

$$\begin{array}{r} 1980 \\ 40 \cdot 20 \\ + 8 \cdot 12 \\ \hline 896 \end{array}$$

$$\begin{array}{r} 2024 \\ 40 \cdot 40 \\ + 8 \cdot 14 \\ \hline 1712 \end{array}$$

$$1950 \rightarrow 1980: \frac{896 - 480}{480} = 86.6\%$$

$$1980 \rightarrow 2024: \frac{1712 - 896}{896} = 91.07\%$$

e) The growth figure decreased both times. Although home activities went up, the rate they increased wasn't proportional to market activities